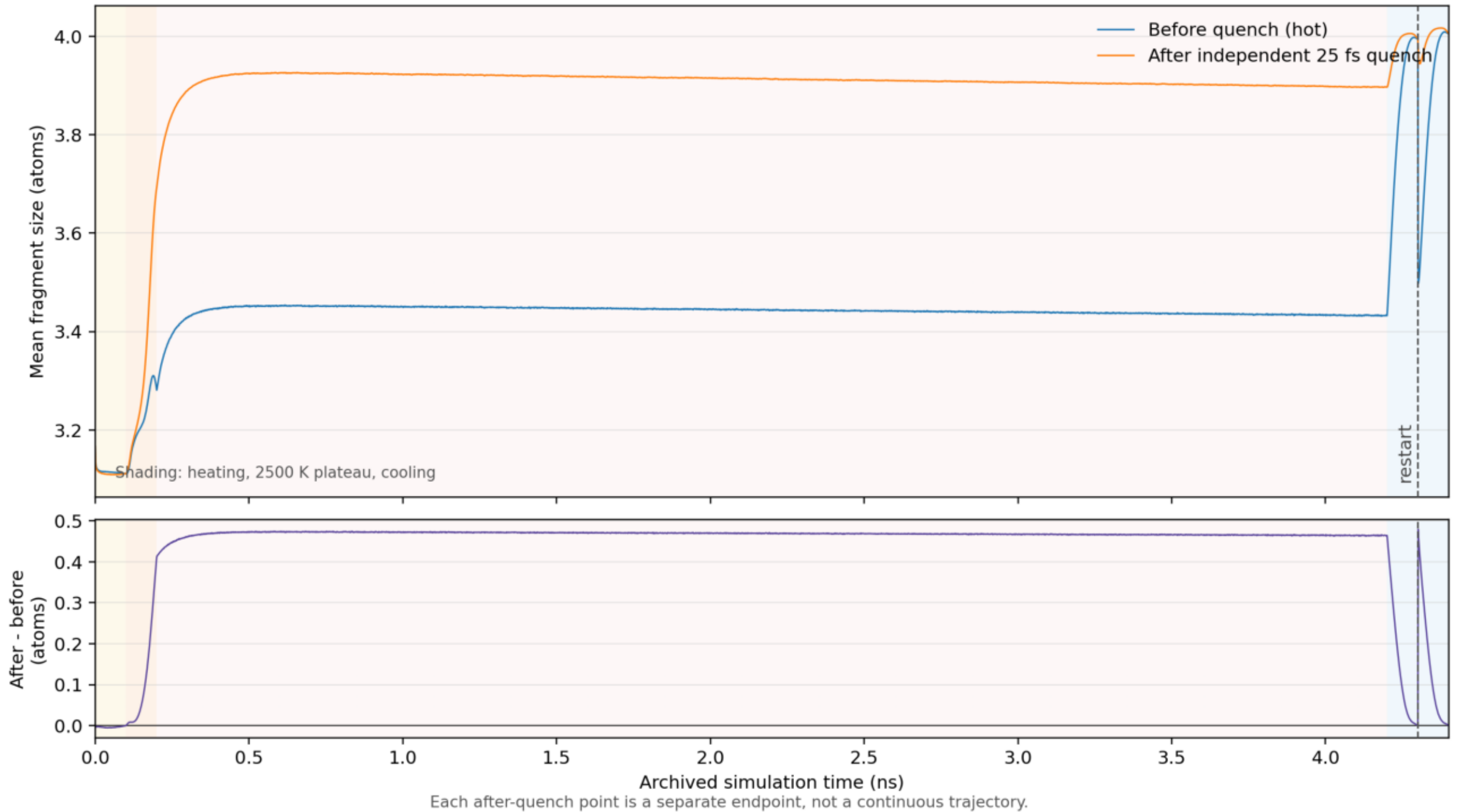


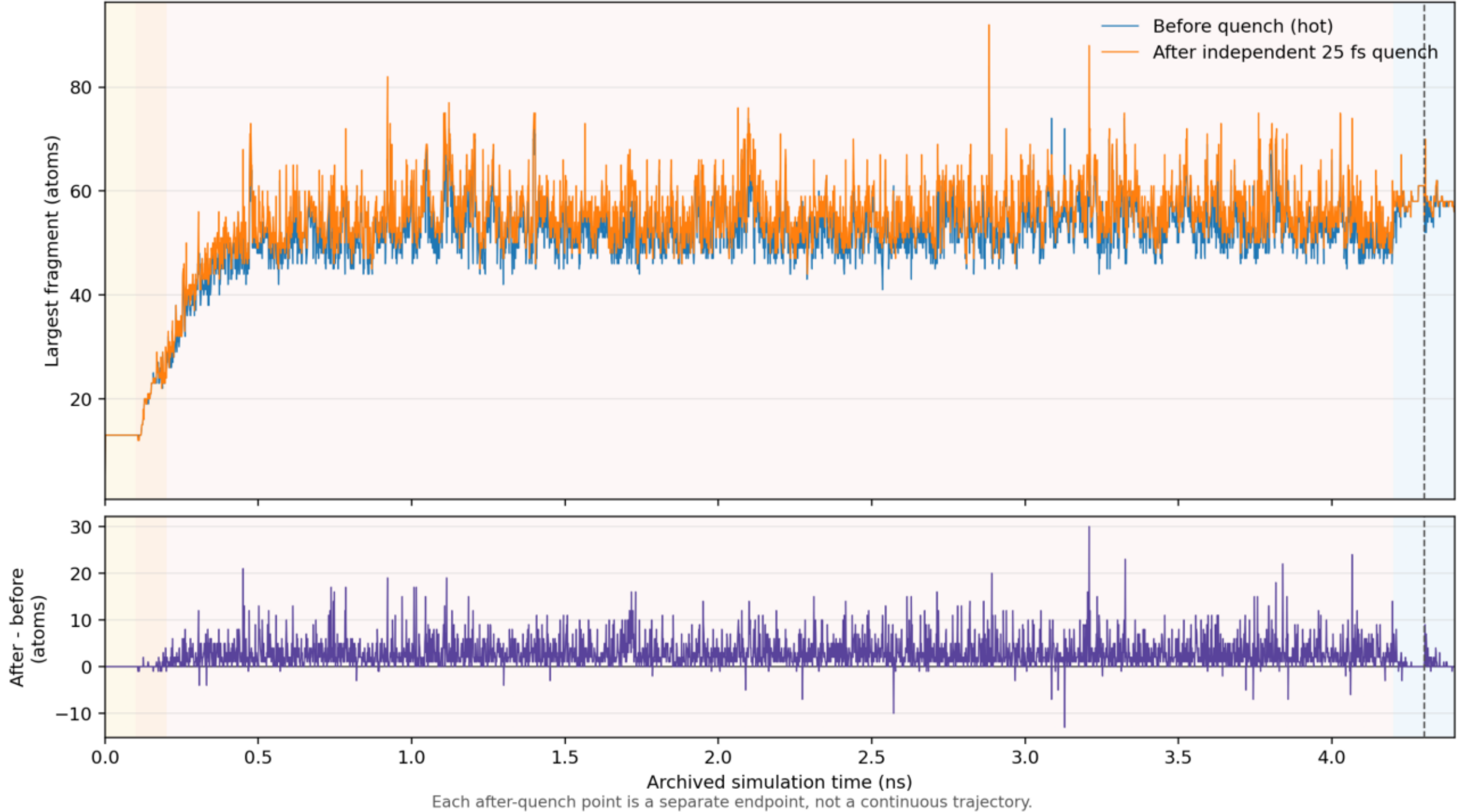
F12: Mean size of every connected fragment

Mean size of all connected fragments across 3,521 paired frames
cohort means: 3.448 before, 3.882 after; after is higher in 3,439 frames



F13: Largest connected fragment

Largest connected fragment across 3,521 paired frames
means: 50.81 before, 53.51 after; after is larger in 2,812 frames

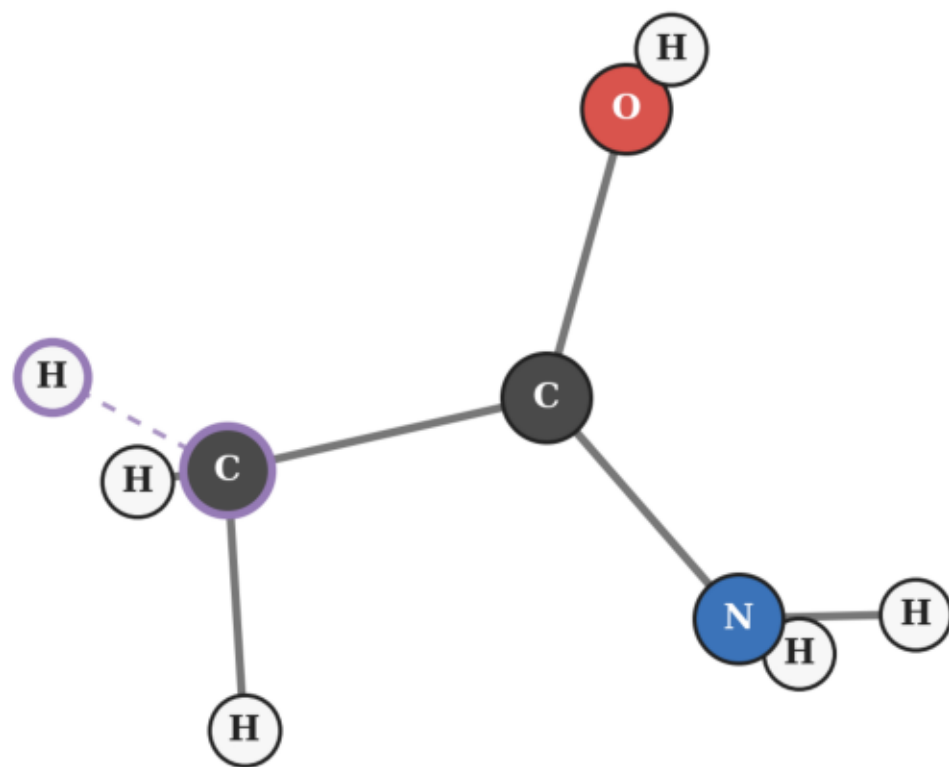


F14: Stable-ID ten-atom connectivity example

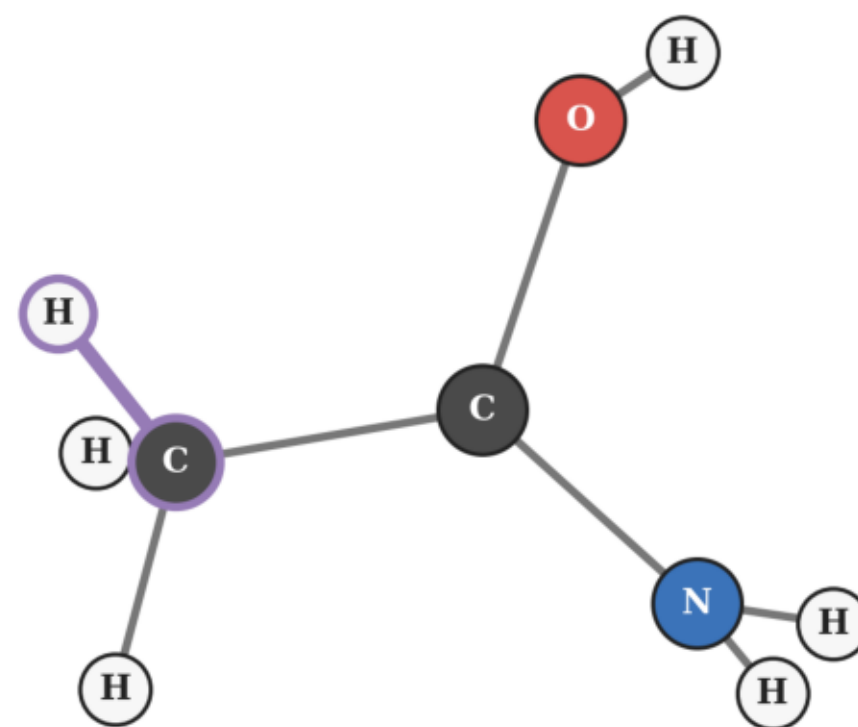
Deterministic frame-447 candidate: identical stable atom IDs before and after quench

Before quench: 9-atom component + isolated H

After 25 fs quench: one 10-atom component



Dashed purple: connection absent in hot endpoint



Solid purple: added cutoff-based C-H connection

Connectivity uses the frozen distance thresholds; it does not assign bond order or establish a reaction mechanism.